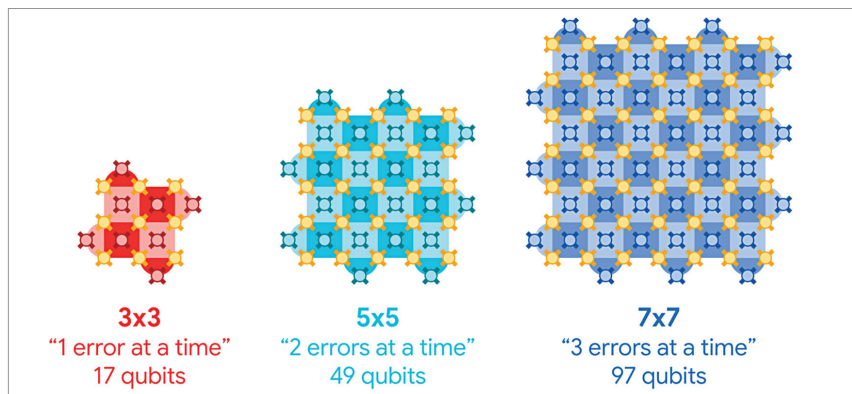


quantum computer, or will it level out at some point? To figure this out, researchers usually build and run repetition codes, a time-tested and efficient type of error-correction that focuses only on bit-flip errors instead of all quantum errors. When running repetition codes on Willow, they were able to realise nearly 10 billion cycles of error correction without seeing an error. However, when they tried to push the encoded error rate lower by increasing the size of the code further, it did not work. This is one of the immediate challenges the team is working on.

The next problem to be tackled concerns the speed of operation. The team has demonstrated the ability to decode measurement information in real time, but decoders can sometimes slow down operations. They have measured a decoder delay of 50 to 100µs on Willow and expect it to go up as the lattice size increases. They are working on bringing this down.

“At current physical error rates, we might need more than a thousand physical qubits per surface code grid to realise relatively modest encoded error rates of  $10^{-6}$ . Furthermore, all of this was accomplished on a 105-qubit processor; can we achieve the same performance on a 1000-qubit processor? What about a million-qubit processor? The engineering challenge ahead of us is immense. At the same time, progress has been staggering, and the



Surface code logical qubits of increasing sizes, each able to correct more errors than the last. The encoded quantum state is stored on the array of data qubits (gold). Measure qubits (red, cyan, and blue) check for errors on the neighbouring data qubits (Image courtesy: Google Quantum AI)

improvement offered by quantum error correction is exponential. We have seen a 20x increase in encoded performance since last year—how many more 20x steps until we can run large-scale quantum algorithms? Maybe fewer than you think,” write Newman and Satzinger.

### Microsoft invents ‘transistor’ for the quantum age

In mid-February came Microsoft’s Majorana 1, a quantum processing unit built using a revolutionary material and new architecture.

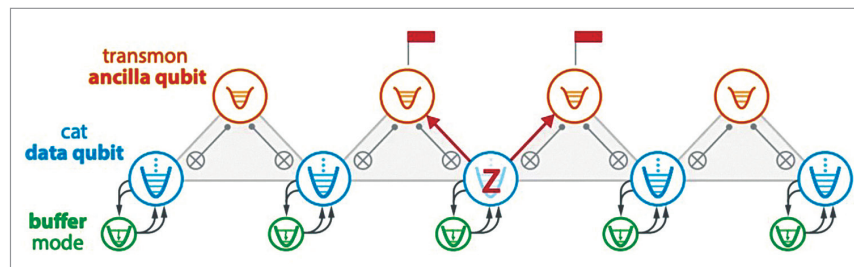
**Revolutionary material.** Many years ago, a team at Microsoft decided to rethink quantum computing at the fundamental level—by creating a new qubit from the ground up! “We took a step back and said ‘OK, let us invent the transistor for the quantum age. What properties does it need to have?’,” says Chetan Nayak, Microsoft

technical fellow, in a blog post introducing the new quantum chip.

Their quest for a stable qubit led them to studies by the mathematician Ettore Majorana, who, more than a century ago, documented a theoretical quasi-particle called the Majorana particle, which could exist in a new state called the topological state. The problem was that these Majoranas did not exist in nature and had never been seen or made!

Microsoft started from the ground up and developed a revolutionary material stack called the topoconductor, which could create this new topological state of matter. They designed and fabricated gate-defined devices with a combination of indium arsenide (a semiconductor) and aluminium (a superconductor), which, when cooled to near absolute zero and tuned with magnetic fields, formed topological superconducting nanowires with Majorana Zero Modes (MZMs) at the wires’ ends. The researchers then harnessed these MZMs to create a qubit that is fast, small, stable, digitally-controllable, and scalable enough to build a quantum system with a million qubits in a palm-sized form factor!

**Simple approach to measurement.** In the new qubit, the MZMs



Pictorial representation of the logical qubit as implemented in the Ocelot chip (Image courtesy: Amazon Science)